Conic Conformal Geometric Algebra

Completing the Object Algebra

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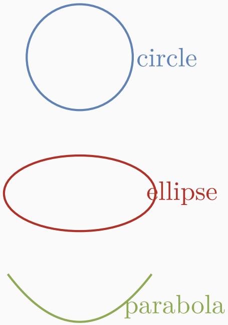
Motivation: from circles to all conics

Conformal GA (CGA) models round objects natively: points, lines, circles, spheres — one isotropic radius.

But an ellipse, hyperbola, parabola, or a tilted conic is *not* round. Classically they need ad-hoc 3×3 matrices, handled *outside* the algebra.

Goal. A single geometric algebra in which every conic is one object — built from points by $\wedge$, intersected by $\vee$, with a clean dual.



CGA reaches only the top one.

From CGA to CCGA

Conformal GA in one slide

Conformal model $\mathbb{R}^{4,1}$, null basis $\mathbf{e}_o, \mathbf{e}_\infty$:

$$\mathbf{e}_o^2 = \mathbf{e}_\infty^2 = 0, \qquad \mathbf{e}_o \cdot \mathbf{e}_\infty = -1.$$

A point (x, y) lifts to a null vector; the metric is Euclidean distance:

$$p = \mathbf{e}_o + x\,\mathbf{e}_1 + y\,\mathbf{e}_2 + \tfrac{1}{2}(x^2 + y^2)\,\mathbf{e}_\infty, \qquad p^2 = 0,$$

$$p \cdot q = -\tfrac{1}{2}\,\mathrm{dist}(p, q)^2.$$

A circle / sphere is the IPNS vector

$$s = p_c - \frac{r^2}{2}\,\mathbf{e}_\infty, \qquad s^2 = r^2.$$

The ceiling

A single isotropic radius $-\frac{r^2}{2}\mathbf{e}_\infty \Rightarrow$ the locus is always a circle.

A general conic

$$Ax^2 + By^2 + Cxy + Dx + Ey + F = 0$$

has six coefficients — the degree-2 Veronese of (x, y). Embed *that.*

CCGA: the algebra

The algebra $\mathbb{R}^{5,3}$

Three conformal copies: signature $\mathbb{R}^{5,3,0}$ (non-degenerate).

$$\underbrace{\mathbf{e}_1, \mathbf{e}_2,}_{+} \qquad \underbrace{\mathbf{e}_{+1}, \mathbf{e}_{+2}, \mathbf{e}_{+3},}_{+} \qquad \underbrace{\mathbf{e}_{-1}, \mathbf{e}_{-2}, \mathbf{e}_{-3}}_{-}.$$

Null basis ($i = 1, 2, 3$):

$$\mathbf{e}_{o_i} = \mathbf{e}_{+i} + \mathbf{e}_{-i}, \qquad \mathbf{e}_{\infty_i} = \tfrac{1}{2}(\mathbf{e}_{-i} - \mathbf{e}_{+i}),$$

$$\mathbf{e}_{o_i}^2 = \mathbf{e}_{\infty_i}^2 = 0, \quad \mathbf{e}_{o_i} \cdot \mathbf{e}_{\infty_i} = -1, \quad \mathbf{e}_1 \cdot \mathbf{e}_1 = \mathbf{e}_2 \cdot \mathbf{e}_2 = 1.$$

Why $\mathbb{R}^{5,3}$ and not a degenerate metric

Non-degenerate $\Rightarrow$ the pseudoscalar I is invertible $\Rightarrow$ dual, meet and join are all clean (unlike PGA).

Special blades and the two gauge bivectors

Two mutually orthogonal conformal pairs — a Euclidean one and a differential one:

$$\mathbf{e}_o = \mathbf{e}_{o_1} + \mathbf{e}_{o_2}, \quad \mathbf{e}_\infty = \tfrac{1}{2}(\mathbf{e}_{\infty_1} + \mathbf{e}_{\infty_2}), \qquad \bar{\mathbf{e}}_o = \mathbf{e}_{o_1} - \mathbf{e}_{o_2}, \quad \bar{\mathbf{e}}_\infty = \tfrac{1}{2}(\mathbf{e}_{\infty_1} - \mathbf{e}_{\infty_2}),$$

$$\mathbf{e}_o \cdot \mathbf{e}_\infty = -1, \qquad \bar{\mathbf{e}}_o \cdot \bar{\mathbf{e}}_\infty = -1, \qquad \{\mathbf{e}_o, \mathbf{e}_\infty\} \perp \{\bar{\mathbf{e}}_o, \bar{\mathbf{e}}_\infty\}.$$

The two gauge bivectors that organise everything:

$$\boxed{I_o^{\triangleright} = \bar{\mathbf{e}}_o \wedge \mathbf{e}_{o_3}} \text{ (origin gauge)} \qquad \boxed{I_\infty^{\triangleright} = (\mathbf{e}_{\infty_1} - \mathbf{e}_{\infty_2}) \wedge \mathbf{e}_{\infty_3}} \text{ (infinity gauge)}.$$

Top blades $I_o = \mathbf{e}_{o_1} \wedge \mathbf{e}_{o_2} \wedge \mathbf{e}_{o_3}$, $I_\infty = \mathbf{e}_{\infty_1} \wedge \mathbf{e}_{\infty_2} \wedge \mathbf{e}_{\infty_3}$, $I_\varepsilon = \mathbf{e}_1 \wedge \mathbf{e}_2$, pseudoscalar $I = I_\varepsilon \wedge I_\infty \wedge I_o$ with $I^2 = -1$.

Dual $A^\star = A\, I^{-1}$, undual $A\, I$; join $A \wedge B$; meet $A \vee B = (A^\star \wedge B^\star)^\star$.

The point: a degree-2 Veronese embedding

$$\boxed{p \;=\; w\,\mathbf{e}_o + x\,\mathbf{e}_1 + y\,\mathbf{e}_2 + \tfrac{x^2}{2}\,\mathbf{e}_{\infty_1} + \tfrac{y^2}{2}\,\mathbf{e}_{\infty_2} + xy\,\mathbf{e}_{\infty_3} \;\pm\; \tfrac{r^2}{2}\,\mathbf{e}_\infty}$$

the lift of $(w,\, x,\, y,\, \frac{x^2}{2},\, \frac{y^2}{2},\, xy)$.

$$p^2 = \pm r^2 \quad (r{=}0:\ \text{null})$$

$$p\cdot\mathbf{e}_\infty = -w$$

$$p\cdot q = -\tfrac{1}{2}\left[(x{-}x')^2 + (y{-}y')^2\right]$$

Euclidean distance, exactly as in CGA.

- $w = 1$: a finite point — a radius r makes it the isotropic member of the conic family;
- $w = 0$: a point with zero homogeneous coordinate — an ideal point (next slide).

One subtlety: two different ideal points

The embedding has three Veronese strata:

$$\underbrace{\mathbf{e}_o}_{\deg 0} + \underbrace{x\mathbf{e}_1 + y\mathbf{e}_2}_{\deg 1} + \underbrace{\tfrac{x^2}{2}\mathbf{e}_{\infty_1} + \tfrac{y^2}{2}\mathbf{e}_{\infty_2} + xy\,\mathbf{e}_{\infty_3}}_{\deg 2}.$$

Zero homogeneous coordinate ($w = 0$, drop $\mathbf{e}_o$), $p \cdot \mathbf{e}_\infty = 0$:

$$p_\infty = x\mathbf{e}_1 + y\mathbf{e}_2 + \tfrac{x^2}{2}\mathbf{e}_{\infty_1} + \tfrac{y^2}{2}\mathbf{e}_{\infty_2} + xy\,\mathbf{e}_{\infty_3}.$$

Keeps the deg-1 part: a deg-1$\oplus$2 *hybrid* that reads as a line.

True ideal point — keeps *only* the top stratum:

$$v_\infty(v) = \lim_{t\to\infty} \frac{p(tv)}{t^2} = \tfrac{v_x^2}{2}\mathbf{e}_{\infty_1} + \tfrac{v_y^2}{2}\mathbf{e}_{\infty_2} + v_x v_y\,\mathbf{e}_{\infty_3}.$$

$\boxed{\deg 0 : \ \mathbf{e}_o}$

$\boxed{\deg 1 : \ \mathbf{e}_1, \mathbf{e}_2}$

$\boxed{\deg 2 : \ \mathbf{e}_\infty\text{-block}} \longrightarrow v_\infty$

Only the deg-2 stratum survives the limit.

Normalization of objects

Points and round objects are projective — λp is the same object. Two ways to pin a representative:

Finite objects — fix the homogeneous coordinate. Since $p \cdot \mathbf{e}_\infty = -w$, divide by the weight:

$$\boxed{\hat{p} = \frac{p}{-\,p \cdot \mathbf{e}_\infty} \qquad \Longrightarrow \qquad \hat{p} \cdot \mathbf{e}_\infty = -1}$$

Scale-invariants — no normalization needed.

$$\operatorname{sign}(s^2) \ \ \text{(reality)}, \qquad\qquad r^2 = \frac{s^2}{(s \cdot \mathbf{e}_\infty)^2} \ \ \text{(invariant radius)}.$$

Ideal objects ($w = 0$, $p \cdot \mathbf{e}_\infty = 0$). The $\mathbf{e}_\infty$-scale degenerates — no finite radius. Use the coordinate-free adaptive norm $\hat{A} = A/\|A\|$, with $\|\cdot\|$ switching from the $\mathbf{e}_\infty$-weight (finite) to the Euclidean direction norm (ideal).

Conics

IPNS: a grade-1 vector *is* a general conic

For q on the conic, $q \cdot s = 0$ reproduces $Ax^2 + By^2 + Cxy + Dx + Ey + F = 0$ with

$$\boxed{s = -2A\,\mathbf{e}_{o_1} - 2B\,\mathbf{e}_{o_2} - C\,\mathbf{e}_{o_3} + D\,\mathbf{e}_1 + E\,\mathbf{e}_2 - \tfrac{F}{2}(\mathbf{e}_{\infty_1} + \mathbf{e}_{\infty_2})}$$

Shape (A, B, C) lives on the origin side $\mathbf{e}_{o_1}, \mathbf{e}_{o_2}, \mathbf{e}_{o_3}$.

Size F lives on the infinity side $\mathbf{e}_\infty$; (D, E) on $\mathbf{e}_1, \mathbf{e}_2$.

$q \cdot s$ never sees $\mathbf{e}_{\infty_3}$ and sees $\mathbf{e}_{\infty_1}, \mathbf{e}_{\infty_2}$ only through their sum: $\mathbf{e}_{\infty_3}$ and $\bar{\mathbf{e}}_\infty$ are gauge-inert (canonical form sets them 0).

OPNS: a grade-7 blade from five points

Five points fix a conic. The clean OPNS object wedges in the origin gauge:

$$\boxed{C = I_o^{\triangleright} \wedge p_1 \wedge p_2 \wedge p_3 \wedge p_4 \wedge p_5} \qquad \text{(grade 7)}.$$

Incidence is a single wedge, dual to the inner-product test:

$$q \wedge C = 0 \quad \Longleftrightarrow \quad q \cdot s = 0, \qquad s = C^{\star} \text{ (grade 1)}.$$

Role of $I_o^{\triangleright}$

Points only span a 6-D subspace; the two directions they cannot reach are $\bar{\mathbf{e}}_o, \mathbf{e}_{o_3}$, whose blade is $I_o^{\triangleright}$. Wedging it supplies those directions and gauge-fixes the dual to land on the clean grade-1 s.

The wedge ladder

Each extra point raises the grade by one; the dual drops by one.

object	OPNS	grade	dual grade
point	p	1	1 (self)
dipole	$p_1 \wedge p_2$	2	6
tripole	$p_1 \wedge p_2 \wedge p_3$	3	5
quadpole	$p_1 \wedge \cdots \wedge p_4$	4	4
pentapole	$p_1 \wedge \cdots \wedge p_5$	5	3 $\left(-\frac{1}{2}\, s \wedge I_\infty^{\triangleright}\right)$
conic	$\cdot \wedge I_o^{\triangleright}$	7	1 $(= s)$

Key point
The pentapole already is the conic as a point locus ($q \wedge P_5 = 0 \Leftrightarrow q$ on the conic). Wedging $I_o^{\triangleright}$ adds *no* geometry — it only cleans the dual from grade 3 to the grade-1 s.

Named conics: one vector s, six numbers

Every named conic is the same s with a specific (A, B, C, D, E, F):

conic	(A, B, C, D, E, F)	type test
circle (c_x, c_y, r)	$(1, 1, 0, -2c_x, -2c_y, c_x^2+c_y^2-r^2)$	$A{=}B$, $C{=}0$, $s^2{=}r^2$
ellipse (a, b)	$(\frac{1}{a^2}, \frac{1}{b^2}, 0, \ldots)$	$\Delta = C^2-4AB < 0$
hyperbola (a, b)	$(\frac{1}{a^2}, -\frac{1}{b^2}, 0, \ldots)$	$\Delta > 0$
parabola $y^2{=}4px$	$(0, 1, 0, -4p, 0, 0)$	$\Delta = 0$
tilted ellipse θ	$C \neq 0$ via $\mathbf{e}_{o_3}$	$\Delta < 0$
line $n_x x+n_y y+d$	$(0, 0, 0, n_x, n_y, d)$	$A{=}B{=}C{=}0$ (no $\mathbf{e}_o$)
line pair $\ell_1\ell_2$	symmetric square of ℓ_1, ℓ_2	$\det M_3 = 0$

A circle is $s = p_c - \frac{r^2}{2}\mathbf{e}_\infty$ — exactly the CGA form. A line is the most degenerate conic: it has *no* origin part.

Closed form of the dipole (point pair)

Parametrise by center p, unit direction v, signed radius r (endpoints $p \pm rv$). The quadratic point map collapses the wedge to

$$\boxed{pp = p_1 \wedge p_2 = -2r\left(C + r^2\, v_\infty\right) \wedge W}$$

$$\begin{aligned} C &= \mathbf{e}_o + p_x\mathbf{e}_1 + p_y\mathbf{e}_2 + \cdots && \text{(center)} \\ W &= v_x\mathbf{e}_1 + v_y\mathbf{e}_2 + \cdots && \text{(tangent d}C[v]\text{)} \\ v_\infty &= \tfrac{v_x^2}{2}\mathbf{e}_{\infty 1} + \tfrac{v_y^2}{2}\mathbf{e}_{\infty 2} + v_x v_y \mathbf{e}_{\infty 3} && \text{(ideal pt of } v\text{)} \end{aligned}$$

Reality from $\operatorname{sign}(r^2)$:

$$\begin{cases} r^2 > 0 & \text{two real pts} \\ r^2 = 0 & \text{tangent / coincident} \\ r^2 < 0 & \text{imaginary pair} \end{cases}$$

$$pp^2 = 4r^4$$

Conic type = incidence with the line at infinity

Build a conic from 3 points + an ideal-point pair B (a line in the plane at infinity):

$$C = p_1 \wedge p_2 \wedge p_3 \wedge B \wedge I_o^{\triangleright}.$$

The type is how B's line meets the conic-at-infinity (the Veronese cone):

B (two ideal points)	line vs ∞-conic	type
$v_\infty(v_1) \wedge v_\infty(v_2)$, real	secant	hyperbola (dirs = asymptotes)
one double direction	tangent	parabola
two imaginary directions	non-secant	ellipse
$I_\infty^{\triangleright}$ (circular points)	non-secant	circle

Reality here is *not* B^2 (infinity is null, $B^2 = 0$): it is the secant / non-secant nature.

Flats and ideal elements — the join units

object	construction	grade
flat point	$p \wedge I_\infty$ (pure position)	4
line (flat)	$p_1 \wedge p_2 \wedge I_\infty$	5
plane (2-flat)	$p_1 \wedge p_2 \wedge p_3 \wedge I_\infty$	6
line at infinity	I_∞	3
conic at infinity	$I_o^{\triangleright} \wedge I_\infty$	5

Composition: flat $\wedge$ flat $= 0$ (repeated I_∞), flat $\wedge$ ideal $= 0$, flat $\wedge$ finite point $\rightarrow$ higher flat.

Flats are for joins; the IPNS line (a degenerate conic) is for meets.

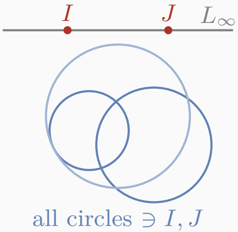
CGA lives inside CCGA

The circular points, and recovering CGA

The infinity gauge is the pair of circular points I, J:

$$I = v_\infty(1, i), \quad J = v_\infty(1, -i), \qquad \boxed{I \wedge J = -\,i\, I_\infty^{\triangleright}}.$$

Every circle passes through I, J; a general ellipse does not.



So wedging $I_\infty^{\triangleright}$ = "be round." It recovers the whole CGA round family inside CCGA (each CGA object of grade k lands at grade $k+2$):

$$\text{round point } p \wedge I_\infty^{\triangleright}, \quad \text{circle } p_1 \wedge p_2 \wedge p_3 \wedge I_\infty^{\triangleright}, \; \ldots$$

Meet, intersections, pencils

Meet grade arithmetic

For blades in general position (pseudoscalar grade 8):

$$\boxed{\mathrm{grade}(A \vee B) = \mathrm{grade}(A) + \mathrm{grade}(B) - 8}$$

Meaningful only when grades sum to ≥ 8; grade 0 is the incidence test.

$A \vee B$	grade	meaning
conic $\vee$ conic	$7{+}7{-}8 = 6$	the 4 Bézout points
conic $\vee$ line at ∞	$7{+}3{-}8 = 2$	the conic's 2 ideal points
conic $\vee$ point	$7{+}1{-}8 = 0$	incidence $q \vee C = 0 \Leftrightarrow q \in C$

$$C_1 \vee C_2 \;\propto\; p_1 \wedge p_2 \wedge p_3 \wedge p_4 \wedge I_o^{\triangleright} \;=\; Q \wedge I_o^{\triangleright} \quad \text{(grade 6).}$$

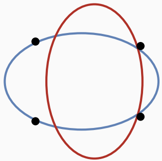
Bézout reduction: the shared-factor principle

Recover the quadpole by contracting out the gauge:

$$Q = (\mathbf{e}_{\infty_3} \wedge \bar{\mathbf{e}}_\infty) \lrcorner (C_1 \vee C_2).$$

Its four points split by reality (sum = 4, Bézout):

$$\{\text{real}\} + \{\text{imaginary}\} + \{\text{ideal}\} = 4.$$



4 Bézout points

$$\boxed{\#\,\text{finite} = 4 - \#\,\text{forced common ideal points}}$$

Two circles share $I, J \Rightarrow$ 2 finite + 2 ideal.

The fast dipole $\pm\sqrt{}$ shortcut is valid *iff* both objects pass through I, J (i.e. both are circles/lines); otherwise use the grade-4 quadpole route.

Pencils of conics

A pencil is the linear family

$$\left\{ \sum_i \lambda_i C_i \right\} \;=\; \text{dual of an } n\text{-blade over } (\mathbf{e}_{o_1}, \mathbf{e}_{o_2}, \mathbf{e}_{o_3}, \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_\infty).$$

All members share the same intersection points.

- a single conic is a 1-pencil; $I_o^{\triangleright}$ is the pencil of *all* conics;
- add a conic: $P \vee C$ (order $+1$);
- add a point: $P \wedge p$ (sub-pencil through p);
- with a right complement A^c: remove a conic $P \wedge C^c$, remove a point $P \vee p^c$.

The complement-based calculus is the next slide's “still to find.”

Properties, read off the algebra

Every conic invariant is an inner product of s

$$A = \tfrac{1}{2} s \cdot \mathbf{e}_{\infty_1}, \quad B = \tfrac{1}{2} s \cdot \mathbf{e}_{\infty_2}, \quad C = s \cdot \mathbf{e}_{\infty_3}, \quad D = s \cdot \mathbf{e}_1, \quad E = s \cdot \mathbf{e}_2, \quad F = s \cdot \mathbf{e}_o.$$

Shape splits into an isotropic part (size) and a deviatoric spin-2 part (tilt):

$$\underbrace{A + B = s \cdot \mathbf{e}_\infty,}_{\text{trace / size}} \qquad \underbrace{(A - B,\ C) = (s \cdot \bar{\mathbf{e}}_\infty,\ s \cdot \mathbf{e}_{\infty_3})}_{\text{anisotropy, angle } = 2\theta}.$$

$$\Delta_2 = AB - \frac{C^2}{4}, \qquad \lambda_\pm = \tfrac{1}{2} s \cdot \mathbf{e}_\infty \pm \tfrac{1}{2} \sqrt{(s \cdot \bar{\mathbf{e}}_\infty)^2 + (s \cdot \mathbf{e}_{\infty_3})^2}, \qquad \theta = \tfrac{1}{2} \operatorname{atan2}(s \cdot \mathbf{e}_{\infty_3},\ s \cdot \bar{\mathbf{e}}_\infty).$$

Center, semi-axes $a_i^2 = -(p_c \cdot s)/\lambda_i$, eccentricity, foci, and asymptotic directions $v_\infty \cdot s = 0$ — all GA; only a final $\surd$ / atan2 is scalar.

Transformations

Versors: one sandwich for every object

$X \mapsto VX\widetilde{V}$ acts uniformly on points, pairs, lines, *and* conics, preserving incidence.

transform	versor
translation	$T = T_x T_y$ (products of $1 - \frac{\tau}{2}\,\mathbf{e}_i \wedge \mathbf{e}_{\infty_j}$)
rotation α	$e^{\alpha E} e^{\alpha K}$, $K = \bar{\mathbf{e}}_o \wedge \mathbf{e}_{\infty 3} - \mathbf{e}_{o3} \wedge \bar{\mathbf{e}}_\infty$, $K^3 = -4K$
scaling s	$\prod_{i=1}^{3}(\cosh u + \sinh u\, \mathbf{e}_{o_i} \wedge \mathbf{e}_{\infty_i})$, $u = \frac{1}{2}\ln s$
reflection	$T(c)R(\theta)V_x\widetilde{R}(\theta)T(-c)$
inversion / transversion	round family only (Möbius $\to$ quartic on general conics)

The rotor turns by 2α — the Veronese symmetric-square action — which is exactly the spin-2 deviatoric pair $(\bar{\mathbf{e}}_\infty, \mathbf{e}_{\infty 3})$ of the previous slide. Shear is non-conformal: no versor exists.

Synthesis and open problems

What is now pure geometric algebra

CCGA $\cong$ GAC (same metric, a basis change $\mathbf{e}_o \leftrightarrow \bar{n}_+$, $\mathbf{e}_\infty \leftrightarrow n_+$, $\mathbf{e}_{o_3} \leftrightarrow \bar{n}_\times, \dots$).

Structural operations expressed as GA products

- objects: points, conics (IPNS grade 1 $\leftrightarrow$ OPNS grade 7), flats, ideal elements;
- incidence, type (Veronese cone), reality;
- discriminants, center, axes, eccentricity, foci, asymptotes — inner products of s;
- intersections (meet), pencils;
- all rigid / conformal motions as versors.

Still to find — the pure-GA gaps

Known GA formulas, not yet unified into the framework:

- the right complement A^c, defined by $A \wedge A^c = I$ — the operator the pencil and orthogonality calculus is built on;
- norm & orthogonality: $\|A\| = \sqrt{(A \wedge A^c)^\star}, \qquad A \perp B \iff A \wedge B^c = 0$;
- center / discriminants as a meet of three dual lines: $\ell_1 \vee \ell_2 \vee \ell_3 = \frac{1}{2}\,\Delta_3\, I_o^\triangleright \wedge I_\infty$;
- the full complement-based pencil calculus $(P \vee C,\ P \wedge p,\ P \wedge C^c,\ P \vee p^c)$.

These are structural and *do* have GA forms — the open work is to fold them into one consistent calculus (via A^c).

Hard problems — the irreducible scalar barrier

GA pushes every structural step into blades. But extracting the actual points is not a GA product:

- tripole $\to$ 3 points needs a cubic (Cardano); quadpole / conic $\vee$ conic $\to$ 4 points needs a quartic (Ferrari). The dipole $\pm\sqrt{}$ has *no* $n \geq 3$ analogue.

Why it is hard, not merely unimplemented

A blade is multilinear (degree 1 in each point), yet the intersection points are the roots of a degree-n polynomial. Radical root-finding for degree ≥ 3 (Abel–Ruffini) is not realizable by the bilinear GA products. GA gives the pencil reduction (resolvent cubic + two $\pm\sqrt{}$ dipoles) cleanly; the final $\sqrt[3]{}/\sqrt{}$ stays an irreducible scalar step.

Still open too: a pure-GA *line-pair factorization*. Objects are always GA multivectors — and we can name exactly where the GA/numeric boundary sits.

Questions?